

Cyclic-Order Laws and Complete Survivor Frontiers in One-Lane Five-Token Dynamics

Terminal-Phase Collapse for Arbitrary Branch Permutations

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2026

This paper (Paper XXXV of the RIME program) studies intrinsic raw terminal incidence in normalized one-lane five-token dynamics. It does not assert typed transfer membership, projectability, recursive return, credit settlement, or a reset bound.

Abstract

Problem. A one-lane branch permutation acts on five source lineages before a marked strict exit. Pair-level Safe-Hit determines which two labels may fuse, but a complete raw frontier must also determine the placement of the three surviving labels and retain the terminal phase used by the same witness.

Approach. We first pass from labelled injections to the 24 oriented cyclic orders of five labels. The branch order law is a diagonal S_5 -invariant relation and is therefore encoded exactly by a subset of the eight double cosets $C_5 \backslash S_5 / C_5$. We then use the globally enabled label-zero return and order-fiber saturation to analyze every formal terminal placement.

Results. The eight-pattern order law generates a lineage-order subgroup $K_a^{\text{ord}} \leq S_5$, and the reachable orders form the left-coset space K_a^{ord} / C_5 . The terminal phase fiber then collapses: a placement whose cyclic order is reachable admits every missing coordinate as a terminal hole, while an unreachable order admits none. Thus every attainable placement has exactly $n - 5$ phase witnesses. For a fused pair F , let $m_a(F) \in \{0, \dots, 6\}$ be the number of compatible linear orders of the three survivors. Then

$$|\mathfrak{S}_a(F)| = m_a(F) \binom{n-2}{3},$$

and the corresponding product parametrization gives the complete normalized raw strict-exit frontier. Equal eight-pattern signatures therefore imply equal phase fibers and equal source-addressed survivor spectra.

Boundary. The theorem is one-lane, unbudgeted, and raw. It does not solve the multi-lane same-witness mobility obstruction, classify shortest terminal words, or cross the typed transfer and projectability boundary.

Keywords. synchronizing automata; single-defect automata; packet lineages; cyclic orders; terminal phase; survivor incidence; double cosets; same-witness descent

Notation Table

Symbol	Meaning
$Q = \mathbb{Z}/n\mathbb{Z}$	labelled circular carrier
$p(q) = q + 1$	fixed full cycle
$E = Q \setminus \{0\}$	normalized one-lane carrier
ε_1	marked collapse $0 \mapsto 1$
$a \in \text{Sym}(E)$	arbitrary branch permutation
T_0	five source lineage labels
$\lambda : T_0 \hookrightarrow E$	labelled normalized state
$\mathcal{L}_a(\lambda_0)$	guarded reachable injection set
$\mathfrak{C}_5$	24 oriented cyclic orders of T_0
R_a	branch-induced relation on $\mathfrak{C}_5$
$\mathcal{P}_5(a)$	eight-class relative-pattern signature
K_a^{ord}	subgroup of S_5 generated by the order law
Ω_a^{reach}	reachable cyclic-order spectrum
F	candidate fused source pair
$R_0 = T_0 \setminus F$	three surviving source labels
$\mu : T_0 \hookrightarrow Q$	formal pre-collapse terminal placement
$\Phi_a(\mu)$	admissible terminal phase fiber
$M_a(F)$	compatible linear orders of the survivors
$m_a(F)$	cardinality of $M_a(F)$
$\mathfrak{S}_a(F)$	source-addressed survivor spectrum

1 Introduction

The raw first-exit frontier of five labelled packets has two layers. The first records which pair of source lineages occupies the terminal kernel pair. The second records where the other three lineages land. The first is a spectator-safe pair-hitting problem. The second is a complete incidence problem and cannot be reconstructed from a separately chosen endpoint.

Paper XXIX classifies the complete canonical raw frontier. Papers XXX and XXXI normalize a general rank- $(n - 1)$ defect into a marked collapse and a free branch permutation. Papers XXXII and XXXIII classify one-lane pair hitting and expose the same-witness obstruction in multiple lanes. Paper XXXIV then shows, within the one-lane dihedral family, that equal fused-pair data can carry different survivor spectra [4–9].

This paper gives the complete one-lane arbitrary-branch answer, beginning with a finite order law. Let

$$\mathfrak{C}_5 = \text{CycOrd}(T_0), \quad |\mathfrak{C}_5| = 24. \quad (1.1)$$

For a branch permutation a , the relation R_a records which cyclic order can be changed into which other order by one application of a . Simultaneous source-label relabelling makes this relation diagonal S_5 -invariant. It is therefore encoded exactly by a subset of the eight relative patterns in

$$C_5 \setminus S_5 / C_5. \quad (1.2)$$

The key step is a terminal-phase collapse. The label-zero return implements a globally, and a finite positive power implements a^{-1} . Hence the reachable injection set is a -invariant. Once a formal terminal placement has a reachable cyclic order, every one of its holes can be used as the deleted coordinate of a genuine terminal witness. Terminal phase is still part of the witness, but it contributes no additional acceptance condition.

This yields the completion theorem of the paper. For each fused pair, the reachable cyclic orders select between zero and six linear orders of the survivor labels. Each selected order may be placed on any three coordinates outside the kernel pair. The frontier is therefore a direct product of an order set and a coordinate subset set.

The three principal results are:

$$\begin{array}{ll}
\textbf{Theorem A:} & R_a \leftrightarrow \mathcal{P}_5(a), \\
& \Omega_a^{\text{reach}} = \text{Reach}_{R_a}(\omega_0) \cong K_a^{\text{ord}}/H; \\
\textbf{Theorem B:} & \Phi_a(\mu) = Q \setminus \mu(T_0) \\
& \iff \text{ord}_Q(\mu) \in \Omega_a^{\text{reach}}; \\
\textbf{Theorem C:} & \mathfrak{S}_a(\lambda_0; F) \cong M_a(F) \times \binom{Q \setminus \{0,1\}}{3}.
\end{array} \tag{1.3}$$

Here K_a^{ord}/H is a left-coset space, not a quotient group, and $\binom{Q \setminus \{0,1\}}{3}$ denotes the set of three-element subsets of $Q \setminus \{0,1\}$.

In Theorem B the phase fiber is empty when the displayed order condition fails. Theorem C is the complete one-lane survivor-frontier classification: all branch-dependent information is carried by the finite set $M_a(F)$, while the coordinate choice is free.

The argument proceeds as follows:

1. Section 2 states the normalized branch system and the inherited order-fiber saturation theorem.
2. Section 3 proves Theorem A: the 24-state order reduction, eight-pattern compression, and lineage-order subgroup description.
3. Section 4 proves Theorem B: terminal-phase collapse and the same-witness terminal criterion.
4. Section 5 proves Theorem C: the complete survivor-spectrum and raw-frontier classification.
5. Section 6 records controls, scope, and the raw-to-typed firewall.

1.1 Related Work and Novelty Boundary

The Černý conjecture and the general theory of synchronizing automata concern short reset words [3, 16, 17]. Circular, almost-group, and completely reachable automata provide neighboring structural settings [1, 2, 10, 11, 13, 14]. This paper proves neither a reset bound nor a complete-reachability characterization.

Subset synchronization, monotonicity, and dynamic word constraints concern related partial and order-sensitive reachability problems [12, 15, 18]. Here every intermediate step must keep five source-labelled lineages injective, and the output retains the incidence of every source label at one declared strict exit.

Within the RIME line, the new claims are an exact eight-pattern order law for terminal incidence, an all- n terminal-phase collapse for arbitrary one-lane branch permutations, and a complete source-addressed survivor formula. The result extends the dihedral orientation classification of Paper XXXIV but does not reopen its pair-level non-descent theorem.

2 The Normalized One-Lane System

Fix

$$n \geq 6, \quad Q = \mathbb{Z}/n\mathbb{Z}, \quad E = Q \setminus \{0\}, \tag{2.1}$$

with $p(q) = q + 1$. Let

$$\varepsilon_1(0) = 1, \quad \varepsilon_1(q) = q \quad (q \neq 0), \quad (2.2)$$

and fix an arbitrary permutation

$$a \in \text{Sym}(E). \quad (2.3)$$

For each label $r \in Q$, the normalized partial return is

$$\phi_r = \varepsilon_1 p^r a : E \longrightarrow E. \quad (2.4)$$

Let T_0 be a five-element source-label set. A normalized state is an injection $\lambda : T_0 \hookrightarrow E$. The transition $\lambda \xrightarrow{r} \phi_r \lambda$ is defined exactly when ϕ_r is injective on the complete image $\lambda(T_0)$. For a fixed source λ_0 , write

$$\mathcal{L}_a(\lambda_0) \quad (2.5)$$

for the complete set of injections reachable by finite guarded paths. No length budget is imposed.

For an injection λ , let $\text{ord}(\lambda)$ be the oriented cyclic order obtained by reading the five source labels around the punctured lane and deleting holes. Orders are taken modulo cyclic rotation, not reflection.

We use two inherited one-lane results. Paper XXXII, Theorem 4.3 supplies identity-branch order-fiber transitivity, while Paper XXXIV, Lemma 3.1 supplies arbitrary-branch identity-path simulation [6, 8].

Lemma 2.1 (order-fiber saturation).

If $\lambda \in \mathcal{L}_a(\lambda_0)$ and $\text{ord}(\nu) = \text{ord}(\lambda)$, then $\nu \in \mathcal{L}_a(\lambda_0)$.

Proof. By Paper XXXII, Theorem 4.3, identity-branch hole slides are strongly connected on every fixed labelled cyclic-order fiber. Paper XXXIV, Lemma 3.1 simulates such an identity path in the arbitrary branch- a system from the same source to the same target injection. Hence one reachable representative saturates its complete order fiber.

The label-zero return will also be essential:

$$\phi_0 = a. \quad (2.6)$$

It is globally enabled because a is a permutation of E .

3 The Finite Cyclic-Order Law

Let

$$\mathfrak{C}_5 = \text{CycOrd}(T_0), \quad |\mathfrak{C}_5| = 24. \quad (3.1)$$

Define the branch order relation by

$$\boxed{\omega R_a \omega' \iff \exists \lambda : T_0 \hookrightarrow E : \text{ord}(\lambda) = \omega, \quad \text{ord}(a\lambda) = \omega'.} \quad (3.2)$$

The relation need not be a function: the image order may depend on the gaps and phase of a representative of ω .

Put

$$\omega_0 = \text{ord}(\lambda_0), \quad \Omega_a^{\text{reach}} = \{\text{ord}(\lambda) : \lambda \in \mathcal{L}_a(\lambda_0)\}. \quad (3.3)$$

Theorem 3.1 (exact order reachability).

$$\boxed{\Omega_a^{\text{reach}} = \text{Reach}_{R_a}(\omega_0)}. \quad (3.4)$$

Proof. In a guarded return $\varepsilon_1 p^r a$, the factor $\varepsilon_1 p^r$ preserves the hole-deleted cyclic label order. Thus every actual return induces one R_a -edge. Conversely, suppose a reachable order ω has an R_a -edge to ω' . Choose an injection realizing that edge. Lemma 2.1 makes this representative reachable. The globally enabled label-zero return $a_\#$ then realizes the edge by an actual guarded step.

3.1 Eight relative patterns

Let $G = \text{Sym}(T_0) \cong S_5$ act on cyclic orders by simultaneous source-label relabelling. If λ witnesses $\omega R_a \omega'$, then $\lambda \tau^{-1}$ witnesses $(\tau \omega) R_a (\tau \omega')$. Hence R_a is diagonal G -invariant.

Take ω_0 as base order and put

$$H = \text{Stab}_G(\omega_0) \cong C_5. \quad (3.5)$$

Then $\mathfrak{C}_5 \cong G/H$, and the diagonal orbitals on $\mathfrak{C}_5^2$ are indexed by $H \backslash G/H$. If $\omega = g\omega_0$ and $\omega' = g'\omega_0$, define

$$[\omega, \omega']_{\text{rel}} = Hg^{-1}g'H. \quad (3.6)$$

The branch signature is

$$\boxed{\mathcal{P}_5(a) = \{[\omega, \omega']_{\text{rel}} : \omega R_a \omega'\} \subseteq H \backslash G/H}. \quad (3.7)$$

Proposition 3.2 (exact pattern reconstruction).

$$\boxed{\omega R_a \omega' \iff [\omega, \omega']_{\text{rel}} \in \mathcal{P}_5(a)}. \quad (3.8)$$

Proof. Each member of $\mathcal{P}_5(a)$ names one diagonal G -orbit. Diagonal invariance includes the complete orbital once any one of its edges is present.

There are eight relative pattern classes. Burnside's lemma gives

$$|H \backslash G/H| = \frac{1}{5}(24 + 4 \cdot 4) = 8. \quad (3.9)$$

Thus $\mathcal{P}_5(a)$ is an exact eight-bit description of the edge law, after any ordering of the eight classes. It does not replace the 24 vertices on which reachability is computed.

3.2 The lineage-order subgroup

For each double coset $D \in \mathcal{P}_5(a)$ choose one representative $g_D \in G$ and define

$$\boxed{K_a^{\text{ord}} = \langle H, g_D : D \in \mathcal{P}_5(a) \rangle \leq S_5}. \quad (3.10)$$

The subgroup is independent of the representatives because it already contains H .

Proposition 3.3 (order subgroup description).

With the fixed left-coset action convention, the reachable-order set is the left-coset space

$$\boxed{\Omega_a^{\text{reach}} \cong K_a^{\text{ord}}/H}. \quad (3.11)$$

Proof. Proposition 3.2 identifies R_a with the union of the orbitals indexed by $Hg_D H$. Starting from H , an edge appends an element of one such double coset. The reachable cosets

therefore lie in K_a^{ord}/H . Conversely each representative edge is present, while left and right factors in H only choose coset representatives. Repeated edges generate the finite semigroup generated by the g_D and H , which is the subgroup K_a^{ord} .

The superscript distinguishes K_a^{ord} , which acts on five source labels, from the physical return group on carrier coordinates in Paper XXX.

Theorem 3.4 (order reduction). *The branch order law satisfies*

$$\boxed{\omega R_a \omega' \iff [\omega, \omega']_{\text{rel}} \in \mathcal{P}_5(a), \quad \Omega_a^{\text{reach}} = \text{Reach}_{R_a}(\omega_0) \cong K_a^{\text{ord}}/H.} \quad (3.12)$$

The final term is the left-coset space of H in K_a^{ord} ; no quotient-group structure is asserted.

Proof. Combine Theorem 3.1, Proposition 3.2, and Proposition 3.3.

4 Terminal-Phase Collapse

Fix a candidate fused pair $F \subset T_0$, $|F| = 2$, and put $R_0 = T_0 \setminus F$. A formal pre-collapse terminal placement is an injection

$$\mu : T_0 \hookrightarrow Q, \quad \mu(F) = \{0, 1\}. \quad (4.1)$$

The terminal exponent must select a hole of μ :

$$0 \notin p^{-u}\mu(T_0) \iff u \notin \mu(T_0). \quad (4.2)$$

For a hole u , the candidate normalized source is

$$\lambda_u = a^{-1}p^{-u}\mu : T_0 \hookrightarrow E. \quad (4.3)$$

Define the phase fiber by

$$\Phi_a(\mu) = \{u \in Q \setminus \mu(T_0) : \lambda_u \in \mathcal{L}_a(\lambda_0)\}. \quad (4.4)$$

Lemma 4.1 (branch invariance).

$$\boxed{a_{\#}\mathcal{L}_a(\lambda_0) = \mathcal{L}_a(\lambda_0).} \quad (4.5)$$

Proof. The globally enabled label-zero return gives the forward inclusion. Since a has finite order, a positive power of the same return realizes a^{-1} and gives the reverse inclusion.

Let $\text{ord}_Q(\mu)$ be the oriented cyclic label order in the full carrier Q .

Theorem 4.2 (all-hole terminal-phase collapse; Theorem B).

For every formal placement μ ,

$$\boxed{\Phi_a(\mu) = \begin{cases} Q \setminus \mu(T_0), & \text{ord}_Q(\mu) \in \Omega_a^{\text{reach}}, \\ \emptyset, & \text{ord}_Q(\mu) \notin \Omega_a^{\text{reach}}. \end{cases}} \quad (4.6)$$

Proof. Fix a hole u . Lemma 4.1 gives

$$a^{-1}p^{-u}\mu \in \mathcal{L}_a \iff p^{-u}\mu \in \mathcal{L}_a. \quad (4.7)$$

The rotation p^{-u} moves an empty coordinate to 0. Deleting that empty coordinate leaves the full cyclic label order unchanged. By Lemma 2.1, the right side of (4.7) holds exactly when $\text{ord}_Q(\mu) \in \Omega_a^{\text{reach}}$. This criterion is independent of the chosen hole.

Corollary 4.3 (phase multiplicity).

Every attainable placement has exactly $n - 5$ terminal phase witnesses. An unattainable placement has none.

Corollary 4.4 (same-witness terminal criterion).

$$\boxed{\mu \text{ is attainable} \iff \text{ord}_Q(\mu) \in \Omega_a^{\text{reach}}.} \quad (4.8)$$

For every selected hole u , the injection $\lambda = a^{-1}p^{-u}\mu$ is reachable and satisfies $p^ua\lambda = \mu$. Thus no endpoint is promoted without an actual source and the same terminal exponent.

Corollary 4.5 (order-law phase descent).

For the same source cyclic order,

$$\mathcal{P}_5(a) = \mathcal{P}_5(b) \implies \Omega_a^{\text{reach}} = \Omega_b^{\text{reach}} \implies \Phi_a = \Phi_b. \quad (4.9)$$

5 Complete Survivor and Frontier Classification

Choose a temporary naming $F = \{f_0, f_1\}$ and let $R_0 = \{r_1, r_2, r_3\}$. A linear order (s_1, s_2, s_3) of R_0 is compatible with F if at least one of

$$[f_0, f_1, s_1, s_2, s_3], \quad [f_1, f_0, s_1, s_2, s_3] \quad (5.1)$$

belongs to Ω_a^{reach} . Let $M_a(F)$ be the set of compatible orders and put

$$m_a(F) = |M_a(F)|, \quad 0 \leq m_a(F) \leq 6. \quad (5.2)$$

The definition is independent of the temporary naming of the two elements of F .

Let $\mathfrak{S}_a(\lambda_0; F)$ be the set of source-addressed survivor injections obtained from actual terminal witnesses over F .

Theorem 5.1 (complete one-lane survivor frontier).

$$\boxed{\mathfrak{S}_a(\lambda_0; F) \cong M_a(F) \times \binom{Q \setminus \{0, 1\}}{3}.} \quad (5.3)$$

Consequently,

$$\boxed{|\mathfrak{S}_a(\lambda_0; F)| = m_a(F) \binom{n-2}{3}.} \quad (5.4)$$

Proof. Let $\{c_1 < c_2 < c_3\}$ be a three-element subset of $Q \setminus \{0, 1\}$ and $(s_1, s_2, s_3) \in M_a(F)$. Map s_i to c_i and map the two labels in F to 0, 1 in an orientation witnessing membership in (5.1). This defines a formal terminal placement whose cyclic order belongs to Ω_a^{reach} . Corollary 4.4 supplies an actual same-witness terminal realization.

Conversely, every terminal witness has a unique survivor coordinate set and a unique linear order of the three survivor labels on that set. Its full cyclic order is reachable, so the survivor order lies in $M_a(F)$. The two constructions are inverse.

Corollary 5.2 (fused-pair criterion).

The pair F is hittable if and only if $m_a(F) > 0$.

Corollary 5.3 (complete normalized raw frontier).

The complete strict-exit incidence over F is obtained from (5.3) by mapping both fused labels to the collapse image 1 and retaining the survivor injection. Hence the relative order law determines the complete one-lane raw frontier.

Corollary 5.4 (complete terminal-incidence invariant).

For branches with the same source cyclic order,

$$\boxed{\mathcal{P}_5(a) = \mathcal{P}_5(b) \implies \mathfrak{S}_a(\lambda_0; F) = \mathfrak{S}_b(\lambda_0; F) \quad \text{for every } F.} \quad (5.5)$$

Proof. *Equal signatures give the same order subgroup and reachable-order spectrum by Theorem A. They therefore give the same $M_a(F)$ for every fused pair. Theorem C then gives identical survivor spectra.*

Complete here means that the signature determines the terminal-incidence frontier. Minimality is not asserted, and the converse implication is not claimed.

5.1 Dihedral recovery

For an orientation-preserving dihedral branch, $m_a(F) = 1$ on the five pairs that are consecutive in the source cyclic order and vanishes otherwise. For an orientation-reversing branch, the corresponding value is 2. Thus Theorem C recovers the Paper XXXIV spectra

$$\binom{n-2}{3} \quad \text{and} \quad 2\binom{n-2}{3}. \quad (5.6)$$

6 Controls, Scope, and Open Boundary

6.1 Six-point exact control

The paper-owned finite package exhausts all $5! = 120$ branch permutations at $n = 6$. It independently computes the 24-state relation, its eight double- coset patterns, guarded injection reachability, terminal witnesses, phase fibers, survivor spectra, and boundary incidence. All branches satisfy the phase-collapse and survivor-count formulas.

At $n = 6$, every injection is a bijection. The eight pattern classes have sizes

$$5, 5, 5, 5, 25, 25, 25, 25 \quad (6.1)$$

and collapse to the five overgroups

$$C_5, \quad D_{10}, \quad F_{20} \cong C_5 \rtimes C_4 \cong \text{AGL}(1, 5), \quad A_5, \quad S_5. \quad (6.2)$$

This no-internal-hole subgroup classification is a finite structural control. The all- n theorems above are proved independently of it.

6.2 What remains open

The result is deliberately restricted in four ways.

1. It is one-lane. The multi-lane capacity and same-witness mobility obstruction of Paper XXXIII remains.
2. It is unbudgeted. No shortest witness or word-length theorem is asserted.
3. It classifies the frontier from $\mathcal{P}_5(a)$ but does not give a simpler carrier-level formula for which branch permutations realize each signature.
4. It is raw. No typed transfer, authorization, ancestry, projectability, or recursive-return statement follows.

The remaining bridge to typed semantics requires two further steps:

$$\begin{array}{c}
 \text{raw complete survivor frontier} \\
 \Downarrow \\
 \text{existing-semantics raw-to-typed bridge} \\
 \Downarrow \\
 \text{same-witness projectability}
 \end{array}
 \tag{6.3}$$

Both arrows remain open under the existing typed semantics.

7 Computational Artifacts

The evidence package is under `experiments/paper35/` in the [RIME repository](#). Paths below are relative to this directory.

Surface	Location	Role
Finite control	<code>order_phase_audit.py</code> , <code>results/</code>	exhaustive six-point consistency control
Proof note	theorem note	supplementary proof provenance
Lean spine	<code>lean/</code>	partial phase, left-coset, and survivor-count formalization
Validation	<code>validation/</code>	source and evidence checks; optional replay
Closure	<code>development-manifest.json</code>	exact 19-artifact inventory

The finite artifacts are consistency controls, not proofs of the all- n theorems or independent computational certificates. The theorem statements and proofs in this manuscript remain authoritative.

A paper-owned Lean formalization machine-checks the all-or-none phase logic, the left-coset type boundary, the survivor-product cardinality, and the forward complete-invariant implication. It consumes the manuscript’s order-fiber saturation and product description; it does not formalize the concrete guarded order graph, the eight-pattern reconstruction, or the finite branch database.

8 Claim Status and Boundary

Claim	Status
Lemma 2.1: order-fiber saturation	inherited theorem, scope preserved
Theorem A: exact pattern/order/subgroup reduction	proved
Theorem 4.2 / Theorem B: all-hole terminal-phase collapse	proved
Theorem C: complete one-lane survivor frontier	proved
Corollary 4.5: equal $\mathcal{P}_5$ gives equal phase fibers	proved
Corollary 5.3: complete normalized raw frontier	proved
Corollary 5.4: $\mathcal{P}_5$ determines the terminal-incidence frontier; minimality and converse open	proved
Phase/left-coset/survivor-product implication spine	machine-checked partial Lean formalization
Six-point branch replay	bounded consistency control
Multi-lane survivor classification	open
Shortest or budgeted terminal witnesses	open
Raw-to-typed transfer bridge	open
Projectable-origin or recursive return	not claimed
Reset bound or settlement theorem	not claimed

9 Conclusion

One-lane terminal incidence is controlled by cyclic-order reachability. The branch law on 24 cyclic orders compresses to eight relative pattern classes, and those classes generate an order subgroup of S_5 . Global branch invertibility then removes terminal phase as an independent acceptance condition: every hole of a reachable placement is valid. The remaining raw frontier is exactly the product of a finite set of survivor orders and a three-coordinate subset.

Thus, within the declared one-lane system,

relative order law $\implies$ all terminal phases $\implies$ complete source-addressed survivor frontier.

The unresolved problems now lie outside that collapse: multi-lane mobility, length-sensitive witnesses, and the existing-semantics bridge from raw incidence to typed transfer and projectability.

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